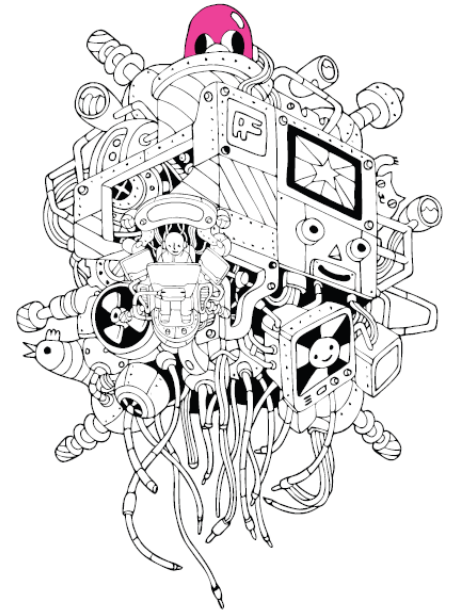


C programming



C Programming:

Chapter 06. Summary of printf

A collection of various cartoon objects arranged in a circular pattern. The objects include a cat, a fish, a planet, a robot, a laptop, a game console, a die, a key, and a character. The objects are drawn in a simple, stylized manner with bold outlines and no shading.

C Programming:

printf function and special characters

```
int main(void)
{
    printf("I like programming \n");
    printf("I love puppy! \n");
    printf("I am so happy \n");
    return 0;
}
```

Execution results

```
I like programming
I love puppy!
I am so happy
```

The printf function prints the string passed as the first argument.

```
printf(" The dog next door said , " Woof ~! Woof ~!!" So cute .");
```

Erroneous printf function call statement

" The dog from the house ne → *Well, this is one string !*
xt door said , " → *What is this ?*
Woof ~! Woof ~! → *This is also a string !*
" It's really cute ."

Double quotes are interpreted as the beginning and end of a string , so if you want to output the double quotes themselves, put \ in front of the double quotes. Let's paste the text !

Background of the birth of special characters

Compiler misunderstanding ?

```
printf(" The dog next door said : \" Woof ~! Woof ~! \" \"It's really cute .");
```

Proper printf function call statement

Types of special characters

Special character	What it means
\a	beep
\b	Backspace
\f	form feed
\n	new line
\r	carriage return
\t	horizontal tab
\v	vertical tab
\'	Output single quote
\"	Output double quote
\?	print question mark
\\	print backslash mark

\f and \v are special characters defined for printer output, not monitor output.

When used for monitor output , strange characters are output !



Format specifications and format characters of the

```
int main(void)
{
    int myAge=12;
    printf("My age is %d years old in decimal and %X years old in hexadecimal. \n", myAge, myAge);
    return 0;
}
```

You can combine the format of the string to be printed using format characters. That is, you can specify the format of the output.

My age is 10 years old in decimal and 16 years old in hexadecimal.

Execution results

```
int main(void)
{
    int num1=7, num2=13;
    printf("%o %#o \n", num1, num1);
    printf("%x %#x \n", num2, num2);
    return 0;
}
```

7 07

d 0xd Execution results

second When inserted , 0 is inserted in front of octal numbers and 0x is inserted in front of hexadecimal numbers .

format text	Output target (data type)	output form
%d	char, short, int	Signed decimal integer
%ld	long	Signed decimal integer
%lld	long long	Signed decimal integer
%u	unsigned int	unsigned decimal integer
%o	unsigned int	unsigned octal integer
%x, %X	unsigned int	unsigned hexadecimal integer
%f	float, double	Decimal floating-point numbers
%Lf	long double	Decimal floating-point numbers
%e, %E	float, double	floating-point real numbers of type e or E
%g, %G	float, double	Select between %f and %e depending on the value
%c	char, short, int	character corresponding to the value
%s	char *	string
%p	void *	address value of pointer

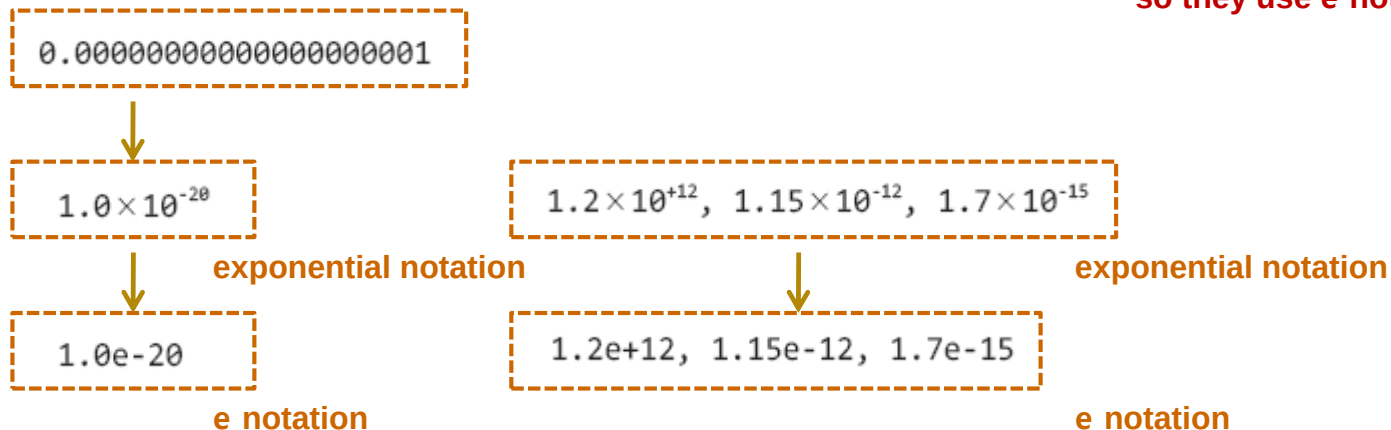
Format characters for printing real numbers : % f, %e

```
int main(void)
{
    printf("%f \n", 0.1234);
    printf("%e \n", 0.1234);    // e 표기법 기반의 출력
    printf("%f \n", 0.12345678);
    printf("%e \n", 0.12345678);    // e 표기법 기반의 출력
    return 0;
}
```

Execution results

```
0.123400
1.234000e-001
0.123457
1.234568e-001
```

Computers cannot represent exponents,
so they use e notation instead .



%g 's real number output and %s 's string output

```
int main(void)
{
    double d1=1.23e-3;    // 0.00123
    double d2=1.23e-4;    // 0.000123
    double d3=1.23e-5;    // 0.0000123
    double d4=1.23e-6;    // 0.00000123

    printf("%g \n", d1);   // %f 스타일 출력
    printf("%g \n", d2);   // %f 스타일 출력
    printf("%g \n", d3);   // %e 스타일 출력
    printf("%g \n", d4);   // %e 스타일 출력
    return 0;
}
```

%g outputs an appropriate format between %f and %e depending on the type of the real number .

The difference between %g and %G is whether the e in the e notation is printed in lowercase or uppercase .

Execution results

```
0.00123
0.000123
1.23e-005
1.23e-006
```

```
int main(void)
{
    printf("%s, %s, %s \n", "AAA", "BBB", "CCC");
    return 0;
}
```

Execution results

```
AAA, BBB, CCC
```

%s of When it comes to string output, let's fully understand it after studying arrays and pointers !

First, let's understand the usage of %s based on examples .



Display neat output by specifying field width

%8d

Set the field width to 8 spaces and output with right alignment.

%-8d

Set the field width to 8 spaces and output with left alignment.

```
int main(void)
{
    printf("%-8s %14s %5s \n", "Name", "Major", "Grade");
    printf("%-85 %14s %5d \n", "Kim Dong-su", "Electronics Engineering", 3);
    printf("%-85 %14s %5d \n", "Lee Eul-su", "Computer Engineering", 2);
    printf("%-85 %14s %5d \n", "Han Seon-yeong", "Art Education", 4);
    return 0;
}
```

Execution results

name	Major department	grade
Dongsu Kim	electronics	3
Eulsu Lee	computer engineering	2
Seonyoung Han	art education	4

The numbers between the format characters indicate the width of the field.

The default is right alignment . Therefore, – is used to mean left alignment.



[illegible]

C Programming:

Defining integer-based input formats

Input format How will you accept it ?

Place of input Where should I save it ?

Entering data

Two pieces of information that must be passed to

%d Data is input in the form of a decimal integer.

%o Data is input in the form of an octal positive integer.

%x Data is input in the form of a hexadecimal positive integer.

similar to the `printf` function,
only changing the output to input.

```
int main(void)
{
    int num1, num2, num3;
    printf("Enter three integers:");
    scanf("%d %o %x", &num1, &num2, &num3);
    printf("Output input integer decimal: ");
    printf("%d %d %d \n", num1, num2, num3);
    return 0;
}
```

Execution results

Input three integers: 12 12 12

Output of input integer in decimal: 12 10 18



Defining input formats based on real numbers

Format characters for inserting float type

In the `printf` function, the format characters `%f`, `%e`, and `%g` each had different meanings. However, the `scanf` function has the same meaning of 'receiving float type data'.

double type long Format characters for inserting

`%lf` double `%f` with **l** added

`%Lf` long double `%f` with **L** added

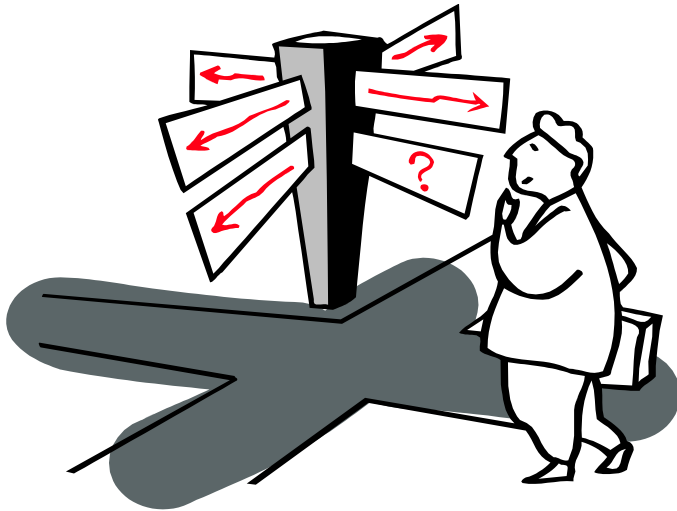
Output data as float, double, and long double `%f`, `%f`, `%Lf`

Input data in float, double, and long double `%f`, `%lf`, `%Lf`

Type	printf	scanf	Description
int	<code>%d</code>	<code>%d</code>	Same
float	<code>%f</code>	<code>%f</code>	Different
double	<code>%f</code>	<code>%lf</code>	Different
long double	<code>%Lf</code>	<code>%Lf</code>	Same

Remember that %s is used for both input and output of strings !

Let's postpone understanding related to this until after studying arrays and pointers !



Chapter 06 This That's it . Do you have any questions ?